Technology Stack & Core Terms

The tools and concepts used to build the existing Resort Management System, grouped by their role.

1. Application foundation

Term	Meaning in this system
Frontend	The browser-facing part of the system: pages, dashboards, forms, and buttons.
React	A library that builds the interface from reusable components.
TypeScript & TSX	TypeScript checks JavaScript code types. TSX is the file format that combines React markup with TypeScript.
Vite	The development server and build tool that turns project code into files for deployment.
Node.js & npm	Tools used to install project packages and run development, test, and build commands.

2. Interface, navigation & visuals

Term	Meaning in this system
Tailwind CSS	Utility classes that style spacing, layouts, colors, and responsive screens.
React Router	Maps URLs to the Admin, Manager, and Receptionist pages.
Radix UI	Accessible building blocks used by interface components such as dialogs and menus.
Recharts	Creates the charts used in dashboards, reports, and analytics.
Lucide React	Provides interface icons such as calendars, search, and edit symbols.

Technology Stack & Core Terms

Data, access, media, and deployment

3. Data, accounts & reliability

Term	Meaning in this system
Firebase Authentication	Signs users in and identifies their accounts.
Cloud Firestore	The database that stores bookings, rooms, payments, users, and other records.
Role-based access control	Routes and actions differ for Administrator, Manager, and Receptionist accounts.
Firestore transaction	Groups related database reads and writes so operations such as walk-in booking stay consistent.
Real-time listener	Watches database records and refreshes connected screens when the records change.

4. Images, hosting & delivery

Term	Meaning in this system
Cloudinary	The existing image-upload service used by some management pages.
Firebase Hosting	Serves the deployed staff website from the generated dist folder.
GitHub Actions	Runs the configured build and deployment workflows when repository events occur.
Production build	The optimized website files Vite creates for hosting.

In one sentence: This is a role-based resort management web application built with React, TypeScript, Vite, Tailwind CSS, and Firebase Authentication, Firestore, and Hosting.